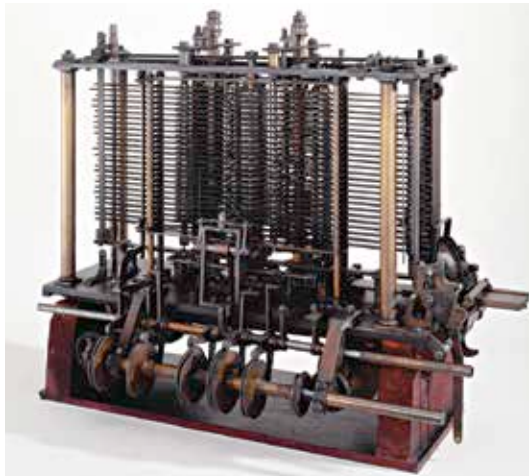


FIRST PROGRAMMER EVER!

We all regard Charles Babbage as the Father of Computing for conceiving the idea of a computer back in the early 1800s. However, his dream of an Analytical Engine would have taken a very



The first general purpose computer

different turn and fight shy of advancing into a modern computer if it wasn't for the efforts of Ada Lovelace – one of Babbage's closest friends and a brilliant mathematician in her own right.

Ada Lovelace is widely regarded as the world's first programmer and a rare visionary. Her most seminal work was done when she translated Charles Babbage's paper on analytical engines that was written by an Italian engineer for a Swiss journal back in 1842. Over a nine-month period, while translating the article, Ada ended up adding her own notes and thoughts on what the engine should do. It exceeded the original piece three times over, having also written a way (program) in which the engine could be used to predict a complex mathematical number sequence. She steered the direction of the Analytical Engine towards concepts that would ultimately evolve it into the general purpose computer that was built in the 20th century.

She introduced the concepts of looping programs and laid down the principles that guided the way letters and symbols could be recognised by future engines with the help of codes (kind of like the conception of ASCII or Unicode). She also envisioned that future machines would be used for more than just mathematical calculations and increasingly for other areas of human interest like

listening to music (no jokes!) and personalisation of computers.

This Enchantress of Numbers, as Babbage used to call Ada, died at the young age of 37, but still managed to leave a lasting impression on the course of computing in its infancy.

QUEEN OF COMPUTING

In the world of computers, if you have to write code and programs in any language, you can't do it without a compiler to help execute it. Compiling a piece of code is, in fact, an essential, fundamental prerequisite to programming and developing rich, complex software. Come to think of it, without compilers the world of computer science would have been at a severe disadvantage, if it weren't for the efforts of a genius named Grace Hopper.

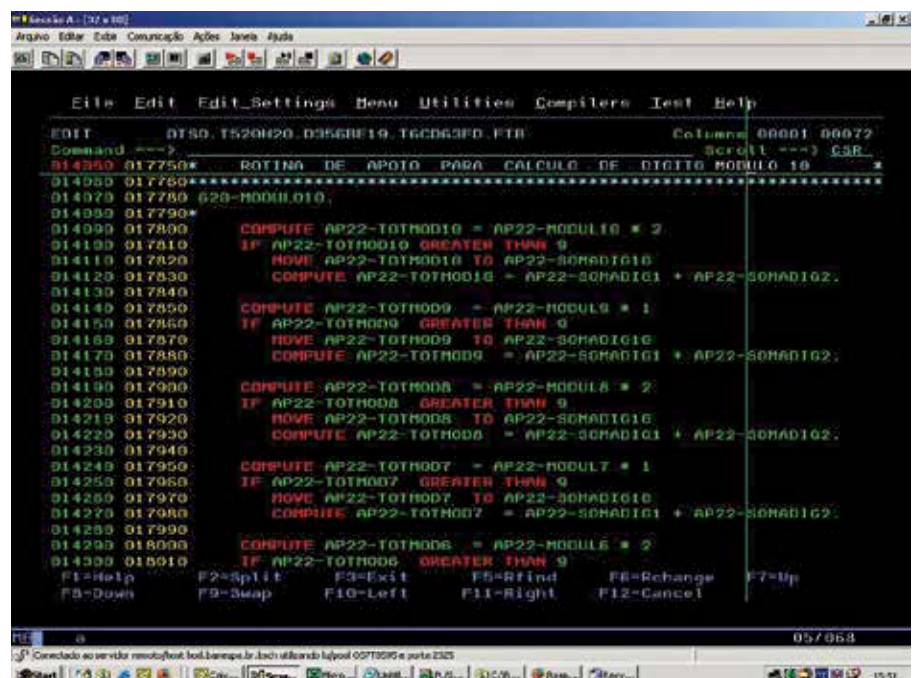
Before joining the US Navy, Admiral Grace Hopper earned a PhD in Mathematics from Yale University as far back as 1934, becoming one of the first women to earn that distinction. During the WWII, Lieutenant Hopper was "tasked with developing a machine [at Harvard] that could make fast, scientific calculations to understand such war-time things as the trajectories of warheads." Her pioneering work birthed one of the first digital computers at that time.

Using FLOW-MATIC, a programming language she designed, Admiral Hopper and her staff were able to make the UNIVAC I and II "understand" twenty statements in English. This was epic at the time because early computers understood English as gibberish, making it nigh impossible for researchers to code in a familiar language. After three years of hard work and perseverance, Hopper published her first compiler paper in 1952. If it were not for her leading effort, COBOL couldn't have been conceived or standardised back in 1959. Every present-day coder and programmer, and in fact anyone involved in any software development role, should tip their hats to Grace Hopper's efforts back in the very infancy of the computing era.

MOTHER OF THE INTERNET

When it comes to the origins of the Internet, we've all heard of Tim Berners-Lee and Vint Cerf, but not many of us have heard of Radia Perlman. They call her the Mother of the Internet, and even though Radia Perlman despises the term we think it's very well justified, as she played an important role in ensuring the Internet worked seamlessly.

A PhD in Computer Science from MIT, Perlman was the first person to solve the problem of file sharing between two computers by her breakthrough 'spanning tree protocol algorithm' or 'STP' in 1985,



COBOL wouldn't have become what it has had n't it been for Grace Hopper